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Can unemployment forecasts based on Google Trends help government design better policies? An investigation based on Spain and Portugal

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Abstract

The Covid-19 pandemic has increased the unemployment issue and accelerated the digital transformation. Real-time data specific to ongoing revolution in applied economic analysis are increasingly demanded to anticipate changes in unemployment to improve decision-making. The aim of this paper is to test whether unemployment rate forecasts based on Google Trends data improve the predictions based only on macroeconomic indicators published with a longer time lag. The research has been carried out at the national level for Spain and Portugal, and the main novelty is the analysis of unemployment rate forecasts at the regional level for Spain using dynamic panel data models to implement the best policies to reduce unemployment. The keywords *unemployment* and *job offers* have been used in each language. The results obtained demonstrate the capacity of Google Trends data associated with *unemployment* to improve the predictions of unemployment rates in Spanish regions. Moreover, predictions based on Google Trends data at national level in Spain and Portugal are significantly more accurate than those based on autoregressive models for both countries.

Keywords: Google Trends; Unemployment; Inflation; Pandemic; Policy

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1. Introduction

Given the severe economic crisis and the sudden strong decline in economic activity, the unemployment variable is currently of particular interest to the general public and for scientists. The improvement of macroeconomic predictions has been a continuous concern of forecasters and policy-makers interested in proposing the best solutions in addressing the social and economic issues. The topic is the subject of current economic debates, marked by skepticism – enhanced by the recent global crisis and Covid-19 pandemic – regarding the predictive capacity of quantitative analysis.

The Covid-19 pandemic was an event beyond anticipated in rational expectations models, with phenomena that are never fully anticipated, and that have profound effects at the time and often for years to come (Stiglitz, 2021). It has exacerbated the unemployment issue and has determined increasing concerns related to a new global economic crisis (Arbolino & Di Caro, 2021). Numerous jobless people contribute to decrease in consumption and investment with direct effects on GDP. Therefore, the unemployment rate is one of the key macroeconomic indicators that could signal the beginning of an economic crisis. In this context, unemployment brings serious welfare issues like social exclusion and inequality. From these points of view, the unemployment evolution presents interest for policy makers, firms, and individuals, since this indicator shows the actual and near-future position of the economy. For these reasons, it is essential to correctly nowcast and forecast the unemployment rate to control its evolution, anticipate trend shifts and design the most suitable pro-employment policies (Mulero & García-Hiernaux, 2021). Moreover, more accurate unemployment forecasts allow the adjustment of GDP forecasts that are necessary to improve the economic decisions in business planning, selection of portfolio in finance, design of economic growth policies.

This paper is a contribution to the ongoing revolution in applied economic analysis. There is an ongoing change in economic analysis from official macroeconomic data released late to big data referring to on-line statistics with real-time data. In this context, this article is an application based on Google Trends tool that provides real-time data referring to job searches to forecast unemployment in Spain and Portugal. The utility of these forecasts is related to policy implications: the advantage to authorities of knowing what goes on in real time to anticipate future evolution of unemployment and the benefit in terms of better decision-making.

In general, the values of unemployment are released late and with revised values in time. A recent example is related to the Great Recession from 2008 when the official data did not anticipate from the very beginning the strength of the economic challenge. Consequently, there is a growing request for real-time data to anticipate the changes in unemployment (Fondeur & Karamé, 2013). In our opinion, this challenge might be addressed if we take into consideration the benefits of the digital economy based on a large amount of Internet data that could improve the forecasting process. The "future" of forecasting should take advantage of technological progress that could contribute to more accurate predictions.

The actual international context is favorable to this approach. Even if the Covid-19 pandemic has enhanced the unemployment issue, it has accelerated the digital transformation. Internet traffic increased about 15–20% in only a few weeks since the beginning of the epidemic (Feldmann et al., 2021) and job searches during the pandemic have increased in all the EU countries and the economic sentiment has deteriorated (van der Wielen & Barrios, 2021). The searches for jobs using the Internet grew by 30% during the lock-down in the EU-27 countries compared to the pre-pandemic period (Caperna, Colagrossi, Geraci, & Mazzarella, 2020).

Therefore, we can conclude that the delay in releasing unemployment rate as a limit of current approaches is overcome by Internet data used to predict unemployment and explain the companies' and labour force behaviour (Simionescu & Zimmermann, 2017). Our hypothesis is that unemployment rate forecasts for Spain and Portugal, countries with high unemployment, based on univariate models could be improved by using search indexes provided by GT. Once digitization of industry and businesses will be achieved as a goal of the European Digital Strategy, the predictive power of internet data will increase, ensuring better unemployment forecasts.

In this context, the main aim of the paper is to check if the unemployment rate forecasts based on Google Trends data improve the predictions based only on macroeconomic indicators that are released quite late. This objective of the research is achieved by conducting an analysis at national and regional level. Besides forecasts for national unemployment rate for Spain and Portugal, this paper brings as a novelty for literature the predictions of regional unemployment rate with quarterly frequency in Spain using Google Trends data. The regional analysis is limited only to Spain because of the lack of data availability for Portugal. Our hypothesis is checked for forecasts at both national and regional level for short horizons. The forecast methods are represented by autoreregressive moving average models for national data and dynamic panel data models for regional data. However, the research might be improved to consider longer horizons. These empirical findings confirm the previous results for Spain and Portugal from other studies and prove as novelty the capacity of Google Trends data to improve unemployment in Spanish regions.

After this introduction, the paper described the previous achievements from literature in this field. The next sections make a detailed description of data, methodology and results. The last section provides conclusions.

2. The ongoing revolution in applied economic analysis

In the last few decades, the revolution in data building and testing was enhanced by growth of the Internet. The big data era is supported by the progress in the registration of social and economic interactions in the digital form (Taylor, Schroeder, & Meyer, 2014). The big data has the capability to transform government, business and other aspects related to economy (Einav & Levin, 2014) and presents advantages and limitations compared to previous approach based on limited macroeconomic or aggregated data.

Our presentation is focused on the debate between applied economic analysis based on limited data and the ongoing revolution in this field to make a comparison between the two perspectives. Then, a particular attention is assigned to this revolution in labour economics, highlighting the benefits of using big data in developing policies and forecasting unemployment rate. After this theoretical framework, the next section of this paper explains the usefulness of using Google Trends to predict unemployment rate in various countries.

The recording of individual behaviour has been a necessity for economic researchers to describe better the economic mechanisms. Before the ongoing revolution in the applied Economics, more limitations of data and methods were observed: insufficient sets of data that were not available in real time, identification of average treatment effect of particular policies neglecting the agents' heterogeneity, weak predictive power, the use of rectangular form for data, analysis based on macroeconomic could conduct to different results compared to microeconomic data.

A first step in overcoming some of these limitations was represented by the use of administrative records, but the main disadvantage was the limited access of public to these administrative data. Consequently, the broad access to these large data becomes a desideratum for governmental agencies and researchers and the seed in developing the concept of 'big data' dealing with extraction, manipulation an analysis of large sets of data. Modern data sets present advantages related to availability in real time and at a larger scale, less structure of data, and new types of variables (Einav & Levin, 2014). The development of new methods to manage this type of data is a challenge for econometric research (Athey & Imbens, 2019).

The traditional microeconomic analysis determines the average treatment effect of specific policies neglecting the heterogeneous character of economic agents that implies non-uniform effect of policy on various agents. This limitation is overcome by large-scale data that allows the construction of econometric models to capture treatment effects and the elaboration of policy forecasts for a large number of subpopulations.

The main applications for big data are related to the knowledge of people behavior in real time and the capability to construct predictive models. For example, companies use big data to manage their business processes and economic indicators and for constructing many types of predictive models. In this case, predictive modelling includes heterogeneity in analysis and in econometric models and naturally achieves stratification in applications related to peoples' behaviour. Authorities might know what goes on in real time to improve decision-making. There are specific applications that convert large sets of unstructured data into predictive scores that could automate existing processes, provide new services and support decision making (Drivas, Giannakopoulos, & Sakas, 2020). Revolution in cultural and educational fields is necessary to take advantage of big data benefits.

Big data could be used to inform policy makers and to improve the government services. The size of the data allows for various strategies to check for results' robustness and hypotheses' validity. Large administrative data are provided by central and local governments in fields like social insurance, education and government spending (Harron et al., 2017), but it remains underused in many cases. According to Card, Chetty, Feldstein, and Saez (2010), administrative data presents a panel data structure (individuals analyzed across time) of high quality. Governments are also interested in supervising the economic activity of private sector. Survey methods have been used before ongoing revolution in data to compute measures for inflation, wages, expenditure, employment and other indicators. The new alternative to survey data is represented by online data. Besides the update in real time, high frequency, granularity and empirical assessment of representativeness, the online data are more reliable in some countries without government survey indicators. However, online data presents some limitations compared to survey data. For example, convenience samples based on Internet data are not always representative (Einav & Levin, 2014).

Another research direction in the ongoing revolution in applied economics refers to the utilization of indirect measures like social media posts and search queries to predict various economic indicators. The most known tools to collect data on people's Internet searches are Google Trends and Twitter. Our paper exploits this direction by using Google Trends to collect data on people's searches on jobs, but more details about this tool are provided in the next section of this paper.

Data analytics are used by government agencies to provide better services and operations, but in some countries personnel and infrastructure are not enough developed to achieve this goal. Another challenge is the use of predictive modeling to ensure better government services.

Besides the importance of the data revolution for government and private sector, this new approach brings opportunities for economic research: granular data ensures better measurement of economic outcomes, allows the use of machine-learning tools and diverse novel research designs. The ongoing revolution still faces more challenges: getting access to data, improvement of data management and programming skills to manage big data, development of creative approaches to analyze the information provided by large-sale data sets.

The penetration of big data in economic fields was slower, because the economic data are usually used for hypothesis testing and not for developing new theories. In this context, analytical methods and big data should be included in traditional economic framework to propose new theoretical models that are close to empirical evidence. Big data in labour economics are among the first applications of this data in social sciences aiming to propose suitable public policies to manage unemployment, inequality, poverty.

The necessity of granular data series to study the interactions between firms and individuals and advance new theories related to labour market was first signaled by Willis (1986) and Rosen (1986). In parallel, administrative data were collected in few Scandinavian countries, the Netherlands and Austria, but without public availability of data. In 1998, the International Symposium on Linked Employer-Employee Data was a step forwards in the construction of large employer-employee microdata, highlighting also issues like data confidentiality and privacy, relevance of these large datasets for policy design, the limitations of usual econometric approaches (Guerrero & Lopez, 2017). The governments passed later from employer-employee microdata to granular administrative records for firms, works and households. In parallel, employer-employee microdata became available to public due to technological progress. The combination of these microeconomic data with economic and demographic characteristics makes the data more reliable for policymaking (Einav & Levin, 2014).

Data analytics represents a powerful tool for authorities and government to understand and aid communities to develop from economic and social point of view. In this context, Google Trends tool is used in this paper to understand the evolution of labour market in Spain and Portugal and to predict the unemployment rate. This type of analysis presents particular interest for government policies.

Planning and forecasting are considered the basis for rational decision making. Through forecasting, information about future economic changes and the impact these changes may have on society can be provided (Granger & Machina, 2006; Martino, 1993; Wieland & Wolters, 2013).

Policymakers need to know about impending changes in the economic environment so they require forecasts to determine the effect of adjustments in the policy instrument, and based on these forecasts they can develop a plan of policy actions. Knowing what is happening in real time is a great advantage for policy makers as they can make better future decisions.

Unemployment is an important macroeconomic variable, it defines the condition of a country's economic equilibrium and is a barrier to social development (Dritsakis & Klazoglou, 2018). It has many social, political and personal implications, since the right to work is one of the basic rights of democracy. Moreover, from an economic point of view, unemployment creates significant inefficiencies and entails significant costs to governments in terms of benefits and retraining programs (Nikolaos, Stergios, Tasos, & Ioannis, 2016).

There are several concrete policies that can be implemented to reduce unemployment. Among them is the partial subsidy of jobs for small and medium-sized enterprises, with the obligation to keep these people employed for a period at least equal to the period of support with public funds and exemption from technical unemployment; the flexibility of employment contracts, from support for part-time employment to the guarantee of full-time contracts, the development of partnership schemes for local employment and co-financing of training for the local labor market, or the university-enterprise partnership, facilitating local governments' collaboration with university institutions for the reinsertion of young people into employment.

3. Google Trends and unemployment analysis

Many studies have shown that the wealth of information from massive data obtained from online search is of great use in studying various social phenomena (Askitas & Zimmermann, 2015; Choi & Varian, 2009; Daas, Puts, Buelens, & van den Hurk, 2015; Einav & Levin, 2014; Johnson et al., 2004; Jun, Yoo, & Choi, 2018; Simionescu, 2020; Van Deursen & Helsper, 2018). This is because people reveal information about their needs and interests when using the Internet (Sherman-Morris, Senkbeil, & Carver, 2011; Zheng, Cai, & Li, 2018).

The most important data source for big data is the internet and the Google Trends index, which shows the intensity of searches on this platform for keywords entered by internet users, is widely used (Campos Vázquez & López-Araiza, 2020).

Google Trends has the ability to define a group of relevant variables and build content from the definition and merging of keywords. Its use has several advantages. Search data is accessible in real time, while data on a target outcome, such as unemployment, is often available with a delay. Official statistics are not only often published with a longer delay: they are also usually subject to substantial revisions (Borup & Schütte, 2020; Eichenauer, Indergand, Martinez, & Sax, 2020). In addition, the daily nature of the search data facilitates a more convincing research design, allowing to track a more detailed pattern of responses given both in the period before and after. Google Trends contains an enormous amount of easily accessible information that is difficult to obtain otherwise. This tool is useful for large search volumes and where the Internet is used frequently. However, Google Trends is limited to an aggregated picture of micro data behaviour and there is insufficient detail on methodology and version failures (Simionescu, 2020).

Google Trends provides temporal data to show the degree of search for a specific keyword in a specific period and a specific location (Carneiro & Mylonakis, 2009; Carrière-Swallow & Labbé, 2013; González-Fernández & González-Velasco, 2018; Nuti et al., 2014). In this way, an index is obtained that is calculated with a representative sample of all Google search data in the indicated time period (Choi & Varian, 2012). This index ranges from 0 to 100 and a value closer to 100 means a higher popularity of the search term in the selected time period and region.

Several studies indicate the importance of using Internet data for forecasting unemployment. Based on the information offered in the WoS of Science database on the most relevant studies on the subject we are analysing in the last decade and considering the "snowball" technique, Table 1 shows the most important aspects of these studies (country for which the study has been carried out, whether it has been done at national or regional level, the specific period, keywords used in the search and the methodology used). Almost all of these studies highlight the usefulness of the Google Trends index in improving unemployment predictions. Most of the researchers work at the national level, with very few at the regional level. The keywords used are almost always directly related to unemployment, although there are papers such as Simionescu, Streimikiene, and Strielkowski (2020) that use the word

Table 1
Reviewing papers over the last decade to predict unemployment with Google Trends.

Authors	Country	Spatial Level	Period	Keywords	Methodology
Milani (2021)	41 countries (Spain, Portugal, among others)	National	January 3-February 6, 2020	Searches related to unemployment, such as unemployment benefits, unemployment insurance, how to apply for unemployment, losing my job	GVAR (Global Vector AutoRegressive)
Mulero and García-Hiernaux (2021)	Spain	National	January 2004-September 2018	Queries related to leading job search applications (e.g., Infojobs, Jobday, LinkedIn); searches related to Spanish unemployment centres (e.g., Employment office, SEPE, Randstad); queries related to standard job searching terms (e.g., Job offers, etc); searches related to those companies that generate the most employment in Spain (e.g., work in Inditex).	SARIMA, PCA and Forward Stepwise Selection (FSS)
Smeekes and Wijler (2021)	Netherlands	National	January 2004–December 2017	Vacancy, resume and unemployment benefits.	Single-equation Penalized Error Correction Selector (SPECS)
Yi, Ning, Chang, and Kou (2021)	U.S.	National	2007–2009	Top 25 Google search terms that are most highly correlated with the term unemployment	PRISM (Penalized Regression with Inferred Seasonality Module)
Borup and Schütte (2020)	U.S.	National	January 2004–February 2019	Jobs and 10 primitive keywords using Google's Keyword Planner	Nonlinear models (such as random forests)

Table 1 (*Continued*)

Authors	Country	Spatial Level	Period	Keywords	Methodology
Campos Vázquez and López-Araiza (2020)	Mexico	National/Regional	2006−2018	Empleo + bolsa de trabajo	AR, LASSO (*Least Absolute Shrinkage and Selection Operator*) method and Random forests
Caperna et al. (2020a, 2020b)	UE27	National	2015–2019	Unemployment	Simple auto-regressive model, Random Forest model
Fajar et al. (2020)	Indonesia	National	January 2004–February 2020	phk (work termination)	ARIMA with Exogenous Variable (ARIMAX)
Fenga and Son-Turan (2020)	Italy	National	2004–2019	Door-to-door (original search term in Italian is porta a porta)	An artificial neural network with a signal processing algorithm of the type autoregressive
Maas (2020)	U.S.	National	January 2004–December 2017	Unemployment	MIDAS regression models
Simionescu (2020)	Romania	Regional	2004−2018	Locuri de muncă, joburi and angajări (jobs and hiring)	Panel data model
Simionescu et al. (2020)	UK	National	July 2016–March 2019	Brexit	Panel data models and multilevel mixed-effects models
Yilmazkuday (2020)	U.S.	National/Regional	January 1 st, 2020–August 24th, 2020	Unemployment	Structural Vector Autoregression (SVAR)
Jung and Hwang (2019)	Korea	National	May 2009–April 2019	18 keywords associated with Korea's youth unemployment rate	SARIMA model
González-Fernández and González-Velasco (2018)	Spain	National	January 2004–November 2017	Desempleo	Regression models, AR model
Naccarato et al. (2018)	Italy	National	January 2004–June 2015	Offerte di lavoro	ARIMA model and a VAR model
Nagao et al. (2019)	U.S.	National	January 2004–June 2018	Jobs, job offer	AR model

Table 1 (*Continued*)

Authors	Country	Spatial Level	Period	Keywords	Methodology
D'Amuri and Marcucci (2017)	U.S.	National/Regional	2007–2014	Jobs	AR model
Lasso and Snijders (2016)	Brasil	National	2004 – 2011	Seguro desemprego (Employment Insurance), *empregos* (jobs) and *décimo terceiro salário* (Social security program), FGTS (Severance Indemnity Fund National), INSS (Institute for Social Security), Unemployment and Social benefits (index)	Linear model with a combination of search levels and first differences
Smith (2016)	UK	National	January 2004–November 2014	Redundancy	MIDAS regression models
Chadwick and Sengül (2015)	Turkey	National	January 2005–January 2012	20 keywords (e.g *Aramak*, which means looking for a job)	Linear regression models and Bayesian Model Averaging
Vicente, López-Menéndez, and Pérez (2015)	Spain	National	January 2004–December 2012	Oferta de trabajo, oferta de empleo	ARIMA models
Su (2014)	China	National	June 2006–November 2012	Unemployment and job search	Basic linear model, Granger Causality Analysis
Barreira et al. (2013)	France, Italy, Spain, Portugal	National	January 2005–August 2013	unemployment (chomage, disoccupazione, desempleo and desemprego)	ARMA
Fondeur and Karamé (2013)	France	National	January 2004–August 2011	Emploi	Univariate, bivariate model, benchmark model Fourier decomposition

Brexit to predict the unemployment rate. In addition, there are two currents, those using a few queries such as González-Fernández and González-Velasco (2018) or Fajar, Prasetyo, Nonalisa, and Wahyudi (2020); and those in favour of using more queries such as Mulero and García-Hiernaux (2021).

Considering the country of application, Naccarato, Falorsi, Loriga, and Pierini (2018) conducted a study for Italy at the national level employing the keyword "offerte di lavoro" (job offer) and using the ARIMA model and the VAR model to forecast the youth unemployment rate. Fenga and Son-Turan (2020) employed the term "porta a porta" ("door to door"), which is typically used by young people to search for low-level jobs that require no specialization to predict youth unemployment in Italy, using an artificial neural network with an autoregressive signal processing algorithm.

Fondeur and Karamé (2013) employed the term "employ", which means "job" but also "employment" in French, to predict unemployment in France and Fajar et al. (2020) employed the abbreviated term used in Indonesia "phk" (work termination) to predict unemployment through an ARIMA with Exogenous Variable (ARIMAX) model, where they showed that during the March to June 2020 period of the COVID-19 pandemic the unemployment rate was expected to increase.

González-Fernández and González-Velasco (2018) employed google data to predict unemployment in Spain through AR model using "unemployment" as a keyword. Mulero and García-Hiernaux (2021) also predicted unemployment in Spain using a SARIMA model. Campos Vázquez and López-Araiza (2020) did the same for Mexico using an autoregressive model and the LASSO method, using "empleo + bolsa de trabajo" as the keyword.

Borup and Schütte (2020) predicted unemployment for the USA, as did Nagao, Takeda, and Tanaka (2019), who used the keywords "jobs", "job offer" under an AR model, although they concluded that the use of Google Trends does not necessarily contribute to improve forecasting accuracy under some preconditions. D'Amuri and Marcucci (2017) also estimated unemployment for the US. These authors concluded that Google-based models predict particularly well at the turning point at the beginning of the Great Recession, although their predictive ability stabilises after this period. Studies were also conducted at the regional level by Yilmazkuday (2020), who used Structural Vector Autoregression (SVAR) to predict unemployment in the US, and Simionescu (2020), who used the terms "locuri de muncă", "joburi and angajări" (jobs and hiring) to predict unemployment in Romania at the regional level, and where he found that Google Trend data improved unemployment models and forecasts.

Our study uses both national and regional approaches. In fact, it is the first study to use the regional approach for Spain.

4. Methodology and data

The methodological approach presented below consists of few key elements where original contributions will be brought: a) Google Trends index (GTI) construction; b) models to explain monthly unemployment in Spain and Portugal at national level; c) models to explain monthly unemployment in Spain at regional level; d) forecasts based on GTI and the selection of the most accurate predictions for national and regional unemployment in the analyzed countries.

a) GTI construction

Considering the mathematical approach, the number of searches for a specific query q is denoted as n(q,c/r,t), where c is the country and r is the region and t is the period, the relative popularity of the query is computed as:

$$RP(q, c/r, t) = \frac{n(q, c/r, t)}{\sum_{q \in Q(l,t)} n(q, c/r, t)} \cdot D_{(n(q,c/r,t)>\tau)}$$

Q(l,t)- set of all queries made during the period t from country c or from region r, $D_{(n(q,c/r,t)>\tau)}$ is a dummy variable that takes the value 1 if the query is popular enough ($n(q, c/r, t) > \tau$) and 0 otherwise. The values obtained are scaled on a range from 0 to 100 considering the proportion of that keyword in the total number of search terms. The GTI is determined as:

$$GTI(q, c/r, t) = \frac{RP(q, c/r, t)}{\max\left\{RP(q, c/r, t)_{t \in 1,2,\ldots,T}\right\}} \cdot 100$$

The GTIs are available from the 1st of January up to 36 h before the search. GTI is zero for queries having low search volume. The repeated searches from the same machine in a very short period are not taken into account, while the filtration of queries with apostrophes and special characters is made.

The GT series were retrieved on August 13th, 2021. The period ranges from January 1st 2004 to December 31st 2020 considering search terms and not topics for "All categories". The countries examined were Spain and Portugal and the data were national and regional just for Spain. Quarterly data for unemployment rate (for 25 years and over) in all the regions are available only for Spain. The searches were individual, not comparisons. The keywords selected were unemployment and jobs in each language: *desempleo* and *ofertas de trabajo* for Spain and *desemprego* and *empregos* for Portugal. Quotes or strings that include "+" were not considered in these keywords.

The regions in Spain are represented by the 17 autonomous regions: Andalusia, Catalonia, Community of Madrid, Valencian Community, Galicia, Castile and León, Basque Country, Castilla-La Mancha, Canary Islands, Region of Murcia, Aragon, Extremadura, Balearic Islands, Asturias, Navarre, Cantabria, and La Rioja.

b) models to explain monthly unemployment in Spain and Portugal at national level

The benchmark is represented by the ARMA model to explain monthly unemployment rate in each country using time series in the period January 2004- June 2021. The period is conditioned by the quarterly GTI availability since 2004. These specifications start from ARMA models described by Box and Jenkins on stationary time series without seasonal factors. An ARMA (p,q) model for unemployment rate is represented as:

$$ur_t = \alpha + \sum_{i=1}^{p} \beta_i ur_{t-i} + \sum_{i=1}^{q} \gamma_i \varepsilon_{t-i} + \varepsilon_t$$

ur_t - unemployment rate at time t

ε_t - error at time t

α - intercept

$\beta_i,\ \gamma_i$ - parameters (coefficients associated to autoregressive component and moving average component of the model)

p,q- order of autoregressive polynomial, respectively moving average polynomial

Table 2
Descriptive statistics for time series (Spain and Portugal) (January 2004-December 2020).

Variable	Mean	Median	Maximum	Minimum	Standard deviation
Unemployment rate in Spain	16.81	16.60	26.40	7.90	5.70
Unemployment rate in Portugal	10.82	9.70	18.20	6.00	3.21
HICP in Spain	96.06	99.04	107.46	78.69	7.37
HICP in Portugal	96.37	98.62	105.50	82.45	6.37
GTIs for *desempleo*	30.70	29.00	100.00	11.00	10.47
GTIs for *desemprego*	47.00	45.00	100.00	15.00	13.40
GTIs for *ofertas de trabajo*	49.23	47.50	100.00	20.00	16.13
GTIs for *empregos*	47.97	44.00	100.00	9.00	20.19

Source: own calculations.

Before estimations, the existence of seasonal factors and the presence of unit root are checked using Augmented Dickey-Fuller test. The series are seasonally adjusted if seasonality is detected, while the time series is transformed in case of non-stationary data to eliminate any unit root.

Two types of alternative models are proposed known as ARMAX models (models that include additional explanatory variables besides the autoregressive and moving average components). We start from this benchmark and include additional variables given by i) GTIs for the mentioned keywords and ii) control variables like harmonized index of consumer prices (HICP) (2015 = 100) which is a measure of inflation.

Seasonally adjusted data from Eurostat were used for unemployment rate that is expressed as percentage of population in the labour force. The unemployed people are defined according to ILO criteria: people aged 15 or over without a job in a particular week that search for a job in the last four weeks and are available to start working the near future up to two weeks (Brandolini, Cipollone, & Viviano, 2006).

Seasonally adjusted data from Eurostat are provided for the Harmonised Index of Consumer Prices (HICP). This indicator shows the change over time in the prices associated to goods and services acquired, paid or used by the households located in euro area.

The average monthly unemployment is higher in Portugal compared to Spain, but the maximum value and the minimum value of this indicator are higher in Spain (see Table 2).

Similar values for average inflation were observed in the two countries. The average GTI for *desempleo* is lower than the values for the rest of the GTIs.

c) models to explain monthly unemployment in Spain at regional level

Given the nature of the data that suppose both temporal and cross-section components, panel data models are constructed for Spanish regions in the period 2004:Q1–2020:Q4 to explain quarterly unemployment rate (*ur*). Only the regions from Spain are chosen because of the data availability. The GTIs are provided for Spanish regions for keywords *desempleo* and *ofertas de trabajo* ($GTI^{desempleo}$, $GTI^{ofertas\ de\ trabajo}$). Inflation rate (*ir*) provided by Spanish Statistical Office is used as control variable in the models.

According to Table 3, the GTIs for the proposed keywords ranged from 0 to 100. The maximum value for unemployment rate (33.9%) was registered in Andalucia in the first quarter of 2013, while the minimum value was observed in Balearic Islands in the last quarter of 2005. Navarre

Table 3
Descriptive statistics for panel data (Spanish regions in the period 2004:Q1–2020:Q4).

Variable	Mean	Standard deviation	Minimum	Maximum
Unemployment rate	14.21	6.68	3.22	33.9
Inflation rate	1.46	1.92	−8.81	10.3
GTI *for desempleo*	14.81	17.48	0	100
GTI for *ofertas de trabajo*	18.51	18.89	0	100

Source: own calculations.

registered the highest inflation in 2011:Q3, while the highest disinflation was observed in Catalonia in 2010:Q3.

Before estimations, cross-sectional dependence, heterogeneity and the existence of unit root is checked. The heterogeneity hypothesis is fulfilled since there are differences between Spanish regions related to unemployment rate values. There are regions with lower unemployment like País Vasco, Castilla y León and Cantabria and regions with higher values (Andalucía, Canarias and Extremadura).

Specific dynamic panel data models on stationary data are considered to explain variation in unemployment rate that is unemployment rate in the first difference ($\Delta ur_{i,t}$):

$$\Delta ur_{i,t} = \alpha_{1i} + \beta_1 \Delta ur_{i,t-1} + \gamma_1 ir_{i,t-1} + \varepsilon_{1i,t}$$

$$\Delta ur_{i,t} = \alpha_{2i} + \beta_2 \Delta ur_{i,t-1} + \gamma_2 ir_{i,t-1} + \delta_1 GTI_{i,t}^{desempleo} + \varepsilon_{2i,t}$$

$$\Delta ur_{i,t} = \alpha_{3i} + \beta_3 \Delta ur_{i,t-1} + \gamma_3 ir_{i,t-1} + \delta_2 GTI_{i,t}^{ofertas\ de\ trabajo} + \varepsilon_{3i,t}$$

$\varepsilon_{1i,t}, \varepsilon_{2i,t}, \varepsilon_{3i,t}$ - errors following normal distribution of null average and constant variance
$\alpha_{1i}, \alpha_{2i}, \alpha_{3i}$ - fixed-effects
$\beta_1, \beta_2, \beta_3, \gamma_1, \gamma_2, \delta_1, \delta_2$ - parameters
i-index for region
t-index for quarter
We assume the independence between fixed-effects and random disturbances (errors).

d) forecasts based on GTI and the selection of the most accurate predictions for national and regional unemployment in the analyzed countries

The models for national and regional data are used to forecast unemployment rate in Spain and Portugal for the next six months. The one-step-ahead forecasts are built on the horizon July 2021–December 2021. These forecasts are assessed using various accuracy measures.

Let us consider ur_t the unemployment rate at time t and the forecast of this variable made at time t for (t + h) denoted by $ur'_{t+h|t}$. We adopt the convention that h = 1 refers to the actual annual forecast. The forecast error is computed as $e'_{t+h|t} = ur_t - ur'_{t+h|t}$.

The forecast accuracy is measured using root mean squared error (RMSE) and Diebold and Mariano (1995) test (DM test). The DM test is applied using a loss function L that is squared error loss (L(e) = e^2) and a benchmark prediction $ur^B_{t|t-h}$ (naïve forecast): $e_t^{DM} = L\left(ur_t - ur'_{t|t-h}\right) - L(ur_t - ur^B_{t|t-h})$. The equal forecast accuracy ($E\left(e_t^{DM}\right) = 0$) is checked using the following

Table 4
The results of KPSS unit root test.

Country	Variable	LM-stat.	Asymptotic critical values
Spain	Unemployment rate	0.672447	0.739000 (1% significance level)
	HICP	1.722947	0.463000 (5% significance level)
	HICP in the first difference	0.370594	0.347000 (10% significance level)
	GTI for *desempleo*	0.418699	
	GTI for *ofertas de trabajo*	0.059407	
Portugal	Unemployment rate	0.426522	0.739000 (1% significance level)
	HICP	1.773728	0.463000 (5% significance level)
	HICP in the first difference	0.397376	0.347000 (10% significance level)
	GTI for *desemprego*	0.481833	
	GTI for *empregos*	0.480476	

Source: own calculations.

model: $e_t^{DM} = a + u_t$, where null hypothesis H_0 states a = 0. The proposed forecast is better than benchmark one if a<0. The RMSE is computed as: $RMSE = \sqrt{\frac{\sum (ur_t - ur'_{t+h|t})^2}{h}}$.

5. Results and discussion

The presentation of the results is made considering the two types of data: national data corresponding to time series and regional data corresponding to panel data. These two types of analyses have the role to show the utility of GT in explaining the unemployment phenomenon at an aggregate and disaggregated level.

5.1. Results at national level

The time series models are built for the period January 2004-December 2020 for Portugal and Spain. Seasonal factors were not identified in the data series associated to GTIs for the keywords proposed. The data for unemployment rate and HICP are already seasonally adjusted. KPSS test is used to check for unit root under the null hypothesis of stationarity as Table 4 shows.

According to Table 4, all the time series are stationary at 1% excepting HICP that is stationary in the first difference ($\Delta HICP_t$). Next, the stationary data sets are used to construct ARMA models. After more estimations, the best models are selected according to Akaike information criterion (AIC) that indicates the model with the lowest AIC from more alternative models. For robustness, ARMAX models are considered (see Table 5).

According to results in Table 5, an AR(3) model for Spain and an AR(2) model for Portugal are the best benchmarks that explain unemployment rate. Only the GTIs associated to keyword *unemployment* in each language had a significant impact on monthly unemployment rate in each country in the period January 2004-December 2020.

The variation in monthly HICP did not have a significant impact on unemployment rate in Portugal and Spain. According to the values of AIC, the best models are those that describe unemployment rate using autoregressive components and GTIs associated to unemployment. The one-step-ahead forecasts for monthly unemployment rate at national level in Spain and

Table 5
The ARMA models to explain monthly unemployment rate in Spain and Portugal (January 2004-December 2020).

Variable	Coefficients for Spain				Coefficients for Portugal			
ur_{t-1}	1.585*	1.606*	1.587*	1.609*	1.356*	1.372*	1.353*	1.376*
ur_{t-2}	−0.311*	−0.342*	−0.315*	−0.348*	−0.360*	−0.377*	0.357	−0.380*
ur_{t-3}	−0.276*	−0.267*	−0.275*	−0.264*	–	–	–	–
GTI_t (unemployment)	–	0.005*	–	0.005*	–	0.003*	–	0.003*
GTI_t (jobs)	–	–	0.0007	–	–	–	0.0009	–
$\Delta HICP_t$	–	–	–	−0.001	–	–	–	0.007
Constant	15.465	15.326	15.414	15.084	8.618	8.549	8.585	8.537
AIC	−0.908	−1.018	−0.903	−1.003	−0.099	−0.199	−0.092	−0.106
Durbin-Watson stat.	1.959	1.899	1.957	1.903	2.188	2.202	2.186	2.201
White test stat.	0.903	0.901	0.907	0.910	0.882	0.887	0.890	0.891
Jarque-Bera stat.	1.12	1.15	1.22	1.16	1.34	1.36	1.35	1.34

Source: own calculations.
Note: * denotes significant coefficient at 5% significance level.

Table 6
Monthly unemployment rate forecasts (%) for Spain and Portugal (January 2021-June 2021).

Month	Forecasts for Spain		Forecasts for Portugal	
	Benchmark model	Model including GTI for unemployment	Benchmark model	Model including GTI for unemployment
January 2021	16.2	16.1	6.7	6.8
February 2021	15.6	15.5	7	7
March 2021	15.5	15.4	6.8	6.7
April 2021	15.1	15.2	6.4	6.5
May 2021	15.6	15.5	7.2	7.1
June 2021	15.3	15.2	7	6.9
RMSE	0.272	0.261	0.270	0.266
DM stat. (p-value in brackets)	37.1 (<0.05)		31.2 (<0.05)	

Source: own calculations.

Portugal are based on benchmark models and those that include GTI for unemployment (see Table 6).

According to RMSE, there are small differences between forecasts based on benchmark models and those based on GTIs for unemployment. However, DM test suggests that the predictions based on GT are significantly more accurate than those based on AR models at 5% significance level for both countries. These results are in line with other studies.

The utility of GT in explaining and predicting monthly unemployment rate in Spain is confirmed by the study of González-Fernández and González-Velasco (2018) who used an AR(1) model as benchmark for the period January 2004-Novemeber 2017. In our case, the AIC for AR(3) was lower than that of AR(1) and this model was chosen in explaining unemployment rate. Moreover, Mulero and García-Hiernaux (2021) also improved the unemployment rate predictions in Spain using GT in the period from January 2004 to September 2018.

For Portugal, Barreira, Godinho, and Melo (2013) showed that GT data have the capacity to improve the monthly unemployment rate forecasts in the period January 2005-August 2013.

Table 7
The results of test for cross-section dependence and unit root.

Variable	CD-test stat.	Breitung test stat. under cross-section dependence (time trend included)	
		no lag	one lag
Unemployment rate	89.73*	0.0871	−0.2021
Unemployment rate in the first difference	–	−5.3224*	−4.0826*
Inflation rate	94.21*	−4.8709*	−3.2948*
GTI for desempleo	12.55*	−8.5083*	−5.4370*
GTI for ofertas de trabajo	7.59*	−8.3509*	−4.9574*

Source: own calculations.
Note: * means p-value less than 0.05.

Table 8
Dynamic panel data models to explain the variation in unemployment rate in Spanish regions (2004:Q1–2020:Q4).

Variable	$\Delta ur_{i,t}$		
$\Delta ur_{i,t-1}$	0.1550*	0.15535*	0.1502*
$ir_{i,t-1}$	0.1222*	0.1235*	0.1256*
$GTI_{i,t}^{ofertas\,de\,trabajo}$	–	−0.0015	–
$GTI_{i,t}^{desempleo}$	–	–	0.0061*
Constant	−0.1188*	−0.0924	−0.2130*
Order	Arellano Bond test for zero autocorrelation in the first-differenced errors (z computed, p-values in brackets)		
1	−3.5575 (0.0004)	−3.5514 (0.0004)	−3.5348 (0.0004)
2	−1.3083 (0.1908)	−1.3078 (0.1910)	−1.3237 (0.1856)
3	−1.5524 (0.1206)	−1.5893 (0.1120)	−1.5517 (0.1207)
4	1.5524 (0.1206)	1.5526 (0.1205)	1.5548 (0.1200)

Source: own calculations.

5.2. *Results at regional level for Spain*

Before estimations for Spainish regions, cross-section dependence and unit root are checked using Pesaran's CD test and Breitung test under the hypothesis of cross-section dependence (see Table 7). The analyzed period refers to 2004:Q1–2020:Q4. For all the variables, the hypothesis of cross-section independence is rejected at 5% significance level. According to Breitung test, the data series are stationary in level, excepting unemployment rate. However, the data set for unemployment rate is stationary in the first difference, at 5% significance level.

Some dynamic panel models are constructed on stationary panel data to explain the variation in unemployment rate in the period 2004:Q1–2020:Q4 (see Table 8). The Arellano–Bover/Blundell–Bond estimator with robust errors is used under the assumption of no correlation between the first differences of instrument variables and the fixed effects. The test for autocorrelation shows no evidence of models' misspecification, since the errors are independend from lag 2 to lag 4 at 5% significance level.

According to results in Table 8, GTI for *ofertas de trabajo* is not relevant in explaining variation in the regional unemployment rate as on the national data. On the other hand, GTI for *desempleo* explains the unemployment rate at regional level in the first difference, even if the influence is low, but a positive one.

Table 9
Forecasts for unemployment rate (%) in Spanish regions (2021:Q1–2021:Q2).

Region	Benchmark model		Model based on GTI for *desempleo*		Accuracy measures	
	Forecasts for 2021:Q1	Forecasts for 2021:Q2	Forecasts for 2021:Q1	Forecasts for 2021:Q2	RMSE for predictions based on benchmark model	RMSE for predictions using model based on GTI for *desempleo*
Andalucía	20.58	20.75	20.37	19.64	0.76	**0.24**
Aragón	10.70	10.54	10.48	9.93	0.38	**0.11**
Principado de Asturias	12.13	13.38	13.06	12.63	1.04	**0.22**
Balears Illes	16.20	18.04	18.33	13.57	3.57	**0.43**
Canarias	23.25	23.03	23.11	22.90	0.09	0.10
Cantabria	10.60	11.28	11.17	11.31	0.52	**0.12**
Castilla y León	10.21	11.75	11.38	11.63	1.06	**0.23**
Castilla - La Mancha	15.73	15.91	15.67	14.75	0.79	**0.26**
Cataluña	11.81	11.18	11.01	10.23	0.72	**0.16**
Comunitat Valenciana	14.77	15.02	14.57	14.80	0.16	0.23
Extremadura	19.43	20.22	20.14	17.50	2.06	**0.07**
Galicia	10.66	12.31	11.96	11.70	1.18	**0.18**
Comunidad de Madrid	12.19	10.68	10.60	10.19	0.99	**0.24**
Región de Murcia	13.55	14.89	14.49	11.63	2.54	**0.26**
Comunidad Foral de Navarra	11.18	9.98	10.33	9.04	0.91	**0.16**
País Vasco	8.60	9.68	9.32	8.27	1.24	**0.13**
La Rioja	9.08	10.91	10.47	9.81	1.51	**0.26**

Source: own calculations.

The benchmark model and that based on GTI for *desempleo* are used to construct one-step-ahead regional forecasts of the unemployment rate in the first two quarters of 2021 in the 17 Spanish regions. These two quarters correspond to a period of pandemic when the use of Internet accelerated compared to pre-pandemic period. RMSE is computed for each region as a measure of forecast accuracy (see Table 9).

The results in Table 9 suggest that forecasts for regional unemployment rate in Spain based on GTI for desempleo are more accurate than those based on the benchmark model that excludes GT data on the horizon 2021:Q1–2021:Q2. In 15 out of 17 regions, RMSE had a lower value for GT forecasts compared to predictions based on benchmark model. For Comunidad Valenciana and Canarias, the benchmark forecasts were more accurate, but without significant differences compared to GT predictions. However, these predictions are limited to a short horizon. In a future study, the forecast accuracy should be assessed for long-run predictions of unemployment rate and using more accuracy measures like mean error, mean absolute error etc. Another limit of this research is the lack of causality between unemployment and GT data, but our purpose is just to provide better forecasts using the econometric models.

All in all, models employing GT data seem to provide more accurate forecasts for Spain and Portugal, but only for a certain key-word. In this case, the GTI associated to the keyword related to unemployment had a significant impact on the unemployment rate, but GTI based on job offers was not relevant. Forecasting the unemployment rate using this keyword in Spain and Portugal at regional level makes it possible to detect, plan and stop any continuous increase in the unemployment rate in a country (Gogas, Papadimitriou, & Sofianos, 2021). From these predictions, strategies and policies can be formulated that are adequate to correct unemployment. In addition, an increase in unemployment usually coincides with a slowdown in growth according to Okun's law, so policies that promote productivity growth should also be implemented (Villaverde & Maza, 2009).

6. Conclusions

The ongoing data revolution is marked by big data and search queries that ensure large sets of data in the economic analysis to help government and companies in understanding and predicting economic indicators. Big data analytics might improve decision-making process in many fields like employment, health, security, productivity and disaster management, but an educational and cultural revolution is also required.

The availability and abundance of data on the Internet makes it possible to explain and forecast macroeconomic indicators such as the unemployment rate. In addition, the Internet makes it possible to observe behavioral trends of individuals as it has become part of many people's daily lives.

Accurately measuring, reducing and controlling the unemployment rate using data in real time provided by Google Trends is one of the main objectives of government. Policymakers usually take unemployment and its variation into account when designing and implementing fiscal and monetary policy. Therefore, it is important to know and predict the unemployment rate in order to be able to implement the right policies and prevent the situation from deteriorating in the future.

The Covid-19 pandemic has exacerbated the problem of unemployment around the world. This concern has been addressed in this paper by taking into account the advantages of the digital economy and the vast amount of data from the Internet that could improve the process of forecasting the unemployment rate to anticipate the changes that occur. Google offers the advantage of obtaining aggregated data in real time with high frequencies based on certain chosen keywords.

It has been analyzed whether unemployment rate forecasts based on Google Trends data for Spain and Portugal improve predictions based on macroeconomic indicators that are usually published with a delay. In addition, the regional unemployment rate has been predicted at a quarterly frequency in Spain using Google Trends data, which is the main novelty of the work.

The keywords selected were *unemployment* and *jobs offers* in each language. The forecast methods are represented by autoregressive moving average models for national data and dynamic panel data models for regional data, with temporal data in the period from January 2004 to June 2021. According to the Akaike information criterion, an AR(3) model for Spain and an AR(2) model for Portugal are the best benchmarks that explain unemployment rate. The models using GT data provide more accurate forecasts for Spain and Portugal, but only for the keyword *unemployment*. The forecasts for regional unemployment rate in Spain based on GTI for *unemployment* are more accurate than those based on the benchmark model that excludes GT data. The keyword *job offers* is not relevant for explaining the variation of the regional and national unemployment rate. Therefore, the governments in Spain and Portugal could have a good image of unemployment evolution if search for GT indexes based on *keyword* unemployment. Moreover, the predictions based on this keyword could help government in designing better policies to ensure protection to unemployed people and to create job opportunities.

Data availability

Data will be made available on request.

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